

Week 1: Titanic Survival Prediction

Machine Learning · Week 1 Task

In this project, I built a machine learning model to predict whether a passenger survived the Titanic disaster. First, I loaded the dataset and cleaned it by filling missing values in the **Age** and **Embarked** columns and removing the **Cabin** column because it had too many missing values. Then, I performed Exploratory Data Analysis (EDA) using different charts to understand how features like gender, passenger class, and age affected survival. After that, I created three new features: **FamilySize**, **IsAlone**, and **Title**, to improve the model's performance. I encoded the categorical data and split the dataset into training and testing sets. Next, I trained two machine learning models: **Logistic Regression** and **Random Forest**. After evaluating both models using **Accuracy**, **Precision**, **Recall**, **F1-score**, and a **Confusion Matrix**, I found that **Random Forest** performed better with an accuracy of **83.80%**, while **Logistic Regression** achieved **81.01%**. Based on these results, I selected **Random Forest** as the final model because it gave the best overall performance for predicting Titanic passenger survival.